Q1.

The names and formulae of five compounds are listed in the table below.

name of compound	formula of compound
ammonia	NH_3
ammonium chloride	NH_4Cl
ammonium sulphate	$(NH_4)_2SO_4$
sodium hydroxide	NaOH
sodium sulphate	Na_2SO_4

(a) Ammonia and sulphuric acid react to give ammonium sulphate, $(NH_4)_2SO_4$.

(i) Balance the equation for this reaction.

$NH_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4$

1 mark

(ii) Complete and balance the symbol equation for the reaction between sodium hydroxide and sulphuric acid.

NaOH + H_2SO_4 → ..... + ...

3 marks

(b) The formula for ammonia is NH_3.
One atom of nitrogen weighs fourteen times as much as one atom of hydrogen.
What is the total mass of hydrogen in 17 g of ammonia?

..... g

1 mark
Maximum 5 marks

Q2.

Galena is an ore of lead. Deposits of galena often contain all of the minerals listed in the table below.

mineral	formula
galena	PbS
calcite	$CaCO_3$
fluorite	CaF_2
zinc blende	ZnS

(a) Give the chemical names of galena and fluorite.

Galena ……

fluorite ……

2 marks

(b) Lead is obtained from galena in two stages.

(i) First the galena is roasted in air. Lead oxide (PbO) and sulphur dioxide are formed as shown in the equation below.
Write in the numbers to balance the equation.

…… PbS + …… O_2 → ……PbO + …… SO_2

1 mark

(ii) The lead oxide is then heated with carbon in a furnace.
Write a balanced symbol equation for the reaction.

……

1 mark

(iii) What is the function of the carbon in this reaction?

……

……

1 mark

(c) Lead can also be obtained from lead oxide by heating it in hydrogen.

The balanced symbol equation for this reaction is:

$$PbO + H_2 \rightarrow Pb + H_2O$$

(i) 112 g of lead oxide, treated in this way, reacted with 1 g of hydrogen to give 104 g of lead. What mass of water was formed in the reaction?

……

……

1 mark

(ii) On heating strongly in air, lead gives lead oxide.
What is the maximum amount of lead oxide you could get from 26 g of lead?

……

……

1 mark
Maximum 7 marks

Q1.

(a) (i) $2NH_3 + H_2SO_4 \rightarrow (NH_4)_2SO_4$

1

(ii) $2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$

one mark is for the formula Na_2SO_4

one mark is for the formula H_20

one mark is for balancing the equation

3

(b) 3

1

[5]

Q2.

(a) lead sulphide

1

calcium fluoride

answers must be in the correct order

1

(b) (i) $2PbS + 3O_2 \rightarrow 2PbO + 2SO_2$

1

(ii) any **one** from

- $PbO + C \rightarrow Pb + CO$
- $2PbO + C \rightarrow 2Pb + CO_2$

1

(iii) it removes the oxygen from the lead oxide

accept 'it removes the oxygen from the lead'
***or** 'it removes the oxygen'*

*accept 'it reduces the lead oxide' **or** 'it is oxidised'*
***or** 'it combines with the oxygen'*

accept 'the carbon displaces the lead in the lead oxide'

1

(c) (i) 9 g

the unit is required for the mark

1

(ii) 28 g

the unit is required for the mark

1

[7]